SCHOOL OF ENGINEERING PROJECT
WEEKLY PROGRESS REPORT

Name : Lee Dong Geon	Admin No : 242558W	Module Code :	Week No : 3	Date of Monday : 23/3/2026

Objectives of the Week :

The main objective for this week was to finalize which LLM to use for the CV screening system. After testing multiple models in the previous weeks, the goal was to evaluate them based on intelligence and speed benchmarks using Artificial Analysis AI benchmark, and decide on the best model to use going forward. The chosen model needed to be reliable, and fast to multiple resumes.

Another objective was to refine the RAG pipeline that I started last week. I had to improve how the system retrieves relevant sections for each candidate, and also build the full LangChain pipeline so that the retrieval and LLM evaluation steps were connected.

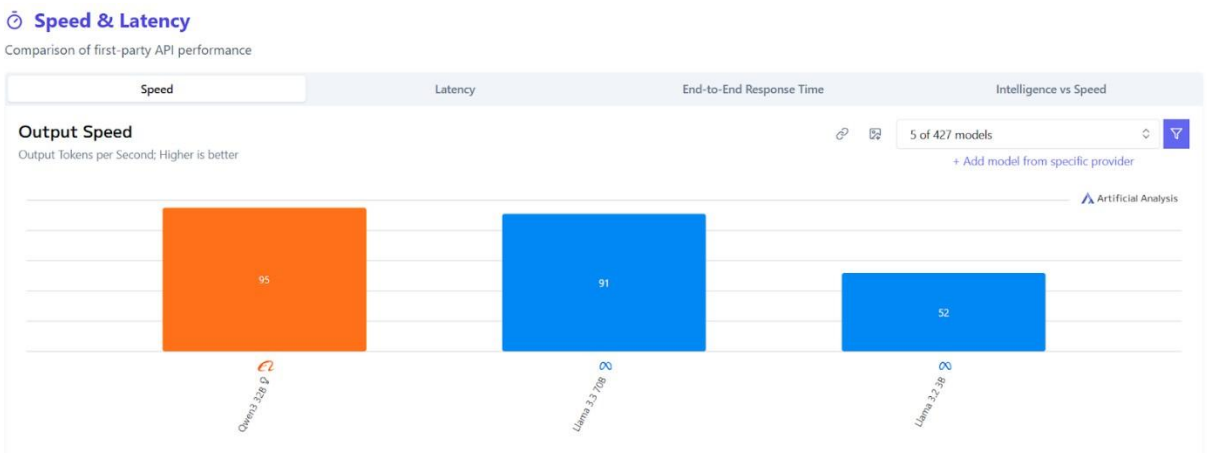
The final objective was to change from the multi model notebook into a one model script, consolidating all the work done so far into one pipeline that I can build upon in future weeks.

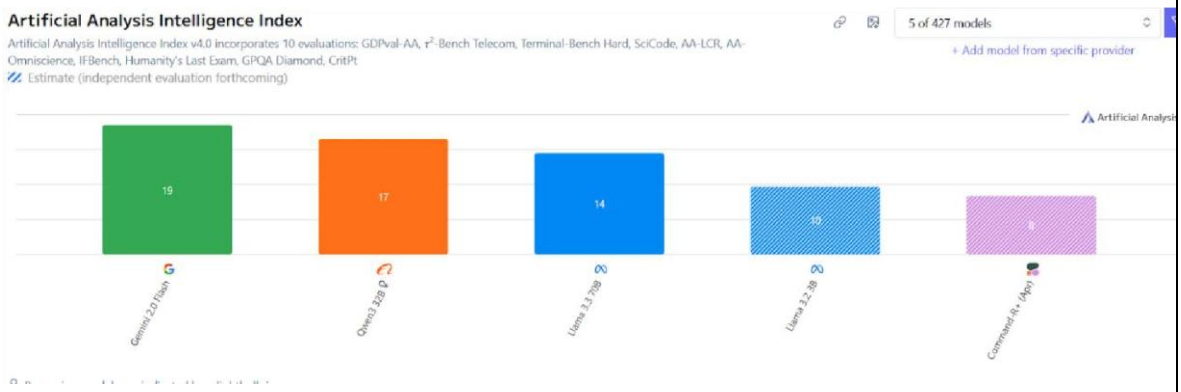
Objectives Achieved and Work Done :

The main focus of this week was selecting the best single LLM to use for the system. From the previous week, five models had been tested: Gemini 2.0 Flash, Qwen3-32B, Llama 3.3 70B, Llama 3.2 3B, and Command-R+. I used the using Artificial Analysis AI benchmark to determine which was the best.

Based on the benchmark, Gemini scored the highest intelligence score of 19 out of 20. However, I eliminated it because it was hitting the daily quota quickly and randomly, which made it unreliable for running all the resumes through. The speed benchmark only included three models: Qwen 3-32B at 95 tokens per second, Llama 3.3 70B at 91 tokens per second, and Llama 3.2 3B at 52 tokens per second. Gemini and Command-R+ did not have speed data available. Out of the three models with speed data, Qwen3-32B was the fastest.

Qwen3-32B was selected as the final model. It had the second highest intelligence score of 17 out of 20, was the fastest model among those with speed data, and ran on the Groq API which was free with generous limits. This was the most practical and reliable choice for the project. I then removed all the other models' information and Api keys that I did not need anymore from my code.





For the RAG part, I improved how FAISS retrieval works. Instead of using one vector store, I had each candidate have their own FAISS vector store This makes sure that the system will only retrieve information from that specific resume, which improves accuracy and avoids mixing with other candidates.

I also adjusted the chunking settings by reducing the size from 400 to 250 characters, with overlap reduced from 80 to 50. I added smaller chunks help the system retrieve more specific sections like skills and experience instead of large mixed content. I also made it so the system retrieved the top 4 most relevant chunks so that there would not be any unnecessary information. However my suitability score went down and two CVs had the same score each time.

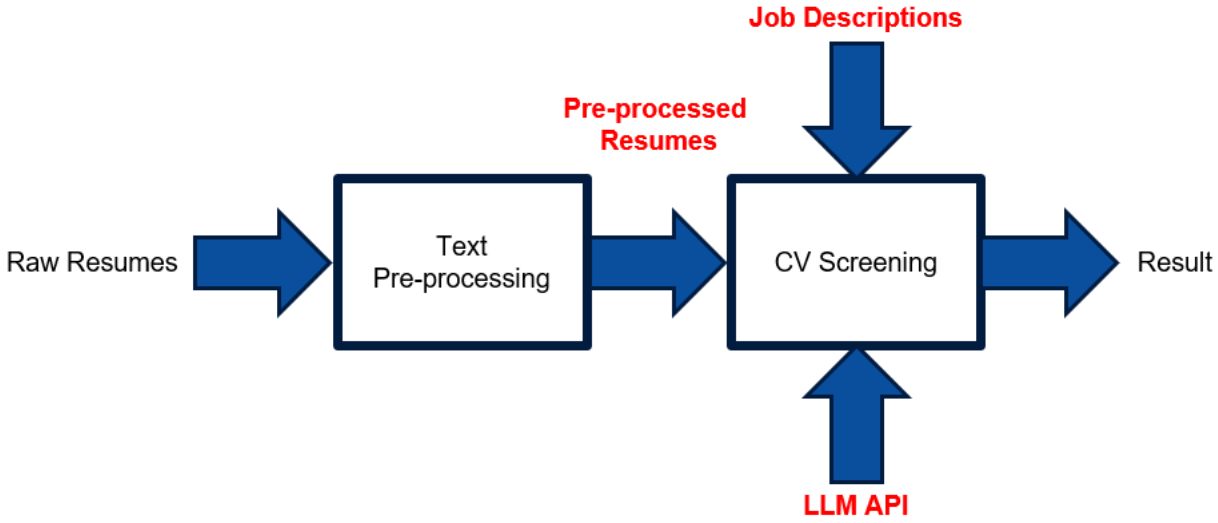
This week, I also built a full LangChain pipeline that connects the FAISS retriever, prompt, Qwen and output into one flow. This makes the system more organised and easier to use.

During testing, RAG gave a tie because the retrieved chunks didn't have enough detail, but Non RAG chose CV 3 since it used the full resume and had more complete information.

This shows that maybe RAG can sometimes miss important details when chunks are too small. Next week, I plan to improve this by increasing chunk size and retrieving more chunks. Overall, the system is now cleaner and more structured.

Task / Type	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Research LLM APIs (Target)												
Research LLM APIs (Actual)												
Select LLM + preprocessing (Target)												
Select LLM + preprocessing (Actual)												
Develop screening (LLM + RAG) (Target)												
Develop screening (LLM + RAG) (Actual)												
Refine screening system (Target)												
Refine screening system (Actual)												
Optimise performance (Target)												
Optimise performance (Actual)												
Mid Presentation (Target)												
Mid Presentation (Actual)												
Develop UI (Target)												
Develop UI (Actual)												
Refine UI (Target)												
Refine UI (Actual)												
Deployment improvements (Target)												
Deployment improvements (Actual)												
Enhance deployment (Target)												
Enhance deployment (Actual)												
Write report (Target)												
Write report (Actual)												
Final submission (Target)												

This Gantt chart shows what I have achieved the past few weeks.



Supervisor's Comment :

3**Appendix 5**

_____ Date/Signature